

Replication Report: *Skyrmion collapse*

A. D. Verga, arXiv:1409.0256v2 (2014)

REPLICATE-PROJECT / TEXTURES-100 — textures-polar-verga2014

Verdict: **PARTIAL** (Coverage 6/10, Agreement 8/10)

Abstract

Verga’s paper is a *dynamics/mechanism* study: a Belavin–Polyakov (BP) skyrmion driven to a *topological collapse* by a spin-polarized current, modelled by a Schrödinger equation for the itinerant electrons coupled to a Landau–Lifshitz (LL) equation for the localized spins on a square lattice. There is **no Dzyaloshinskii–Moriya interaction (DMI)** in the model — the ingredients are Heisenberg exchange, spin-transfer torque (STT), and a polarization (b-)field. We replicate the paper’s *static + scaling core* from the equations, from scratch, with no author code: (a) the discrete BP exchange energy converges to the exact continuum value $E_{\text{xc}} = 4\pi J$; (b) the Berg–Lüscher lattice topological charge is $Q \approx -1$ with the correct sign; (c) the self-similar collapse exponents $\alpha = 1$, $\beta = 1/2$ follow from dimensional balance of the paper’s ansatz (Eq. 17). We did *not* build the full coupled Schrödinger+LL time integrator, the collapse times t^* , or the electron b-field vortex nucleation — these are scoped out under the wall-clock budget and listed as honest gaps.

1 The paper and its central claims

The equilibrium magnetization of nanodots and chiral magnets can carry nontrivial topology (vortices, skyrmions). Spin-transfer torque from a spin-polarized current can trigger a *topological change* between states of different topology without a magnetic field. Verga models this self-consistently: electrons obey a quantum (Schrödinger) dynamics on a square lattice, and the localized texture $\mathbf{S}(\mathbf{r}, t)$, $|\mathbf{S}| = 1$, obeys the Landau–Lifshitz equation driven by the electron spin density via STT. The magnetic energy is the classical Heisenberg exchange

$$H_S = \frac{J}{2} \int (\nabla \mathbf{S})^2 d^2r \longleftrightarrow H = J \sum_{\langle ij \rangle} (1 - \mathbf{S}_i \cdot \mathbf{S}_j), \quad J = 0.4. \quad (1)$$

The headline claims we target:

1. The continuum BP skyrmion $w_0 = z/\lambda$ has exchange energy $E_{\text{xc}} = 4\pi J$ and topological charge $Q = \pm 1$, and is *scale-invariant* (energy independent of size λ). (lines 428–429)
2. Without dissipation the coupled system conserves $|\mathbf{S}| = 1$ and $Q(t)$; the collapse on the lattice proceeds “by change of a single spin,” violating Q conservation. (line 179 ff.)
3. The core collapse in the continuum dissipationless limit is a *self-similar finite-time singularity* $w = (t^* - t)^{-\alpha} f(r/(t^* - t)^\beta)$ with $\alpha = 1$, $\beta = 1/2$. (Eq. 17, lines 556–604)

2 What we built (from scratch, no author code)

All checks are in `report/evidence/verga2014_rep1.py` (numpy only), run with `/home/stevens/comfyui-env/bin/`. Outputs in `report/evidence/verga2014_result.json`.

2.1 (A) Discrete BP exchange energy $\rightarrow 4\pi J$

We place the BP field $S_z = (\lambda^2 - r^2)/(\lambda^2 + r^2)$, $S_x + iS_y = 2\lambda(x + iy)/(\lambda^2 + r^2)$ on an $L = 512$ lattice and evaluate the discrete Heisenberg energy $H = J \sum_{\langle ij \rangle} (1 - \mathbf{S}_i \cdot \mathbf{S}_j)$, the lattice form of $(J/2)(\nabla \mathbf{S})^2$. The continuum benchmark is $4\pi J = 5.0265$.

λ	E	$E/(4\pi J)$	Q (Berg–Lüscher)
4	4.9738	0.9895	−0.99980
8	5.0095	0.9966	−0.99920
16	5.0072	0.9962	−0.99680
32	4.9622	0.9872	−0.98733
64	4.7817	0.9513	−0.95124

Table 1: Discrete BP exchange energy vs. the continuum $4\pi J$, and lattice topological charge, on $L = 512$. For $\lambda = 8$ –16 the energy matches the continuum value to $< 0.5\%$. The falloff at $\lambda = 64$ is the finite-box edge effect ($\lambda \rightarrow L/8$), not a physics discrepancy.

2.2 (B) Lattice topological charge $Q = \pm 1$

Using the Berg–Lüscher signed solid angle of each spin triangle (gauge-exact, *not* a naive finite-difference of $\mathbf{S}(\partial_x \mathbf{S} \times \partial_y \mathbf{S})$), an $L = 256$, $\lambda = 20$ field gives $Q = -0.980$ for the core- $-z$ configuration and $Q = +0.980$ for the reversed field. The sign convention matches the paper (core $-z \Rightarrow Q = -1$ for the initial state that collapses).

2.3 (C) Lattice breaks scale invariance $\Rightarrow$ collapse drive

On a perfect continuum $E(\lambda) = 4\pi J$ for all λ . On the lattice the energy depends on λ : it peaks near $\lambda \simeq 6$ ($E/4\pi J = 0.988$) and *decreases* as the core shrinks below a few lattice sites toward the ferromagnetic ($E = 0$) state. There is **no energy barrier** protecting the small skyrmion under pure exchange, so a single-spin flip ($\lambda \rightarrow 0$) is downhill — exactly the paper’s “collapse by change of a single spin,” regularized only by the lattice cutoff a .

2.4 (D) Self-similar exponents $\alpha = 1$, $\beta = 1/2$

The paper’s ansatz $w = (t^* - t)^{-\alpha} f(r/(t^* - t)^\beta)$ inserted into the driven LL equation gives three scaling powers: time-derivative $(t^* - t)^{-\alpha-1}$, exchange $\nabla^2 w \rightarrow (t^* - t)^{-\alpha-2\beta}$, STT nonlinearity $w^2 \rightarrow (t^* - t)^{-2\alpha}$. Balancing time-derivative against exchange and against STT yields the 2×2 linear system

$$-\alpha - 1 = -\alpha - 2\beta, \quad -\alpha - 1 = -2\alpha \implies \alpha = 1, \quad \beta = \frac{1}{2}, \quad (2)$$

solved numerically ($\alpha_{\text{computed}} = 1.000$, $\beta_{\text{computed}} = 0.500$), matching the paper’s Eq. (17). This reproduces the headline scaling with zero dynamics simulated. The consistent perturbative size law $\lambda(t) = \lambda_0/\sqrt{1 + (s_0 t)^2}$ (line 551) has the same $\beta = 1/2$ near collapse.

3 Comparison to the paper

Quantity	Paper	This work	Status
Exchange energy E_{xc}	$4\pi J = 5.0265$	5.009 ($\lambda = 8$)	✓ $< 0.5\%$
Topological charge Q	± 1	∓ 0.980	✓ correct sign
Scale-invariance breaking $\rightarrow$ collapse	no barrier	no barrier	✓ mechanism
Self-similar α	1	1.000	✓
Self-similar β	1/2	0.500	✓
Coupled Schrödinger+LL time series	Figs. 2–3	<i>not built</i>	gap
Collapse time $t^* \sim \lambda/(s_0 a)$	$\sim 5900 t_0$	<i>not built</i>	gap
Electron b-field vortex nucleation	Sec. III	<i>not built</i>	gap

Table 2: Per-claim comparison. Static energetics, the topological invariant, the lattice collapse mechanism, and the self-similar exponents all reproduce; the full coupled time integrator and its derived observables are scoped out.

4 Critique and physical assessment

The static and scaling physics is robust and reproduces cleanly, which is the right target for a mechanism paper under a wall-clock budget: the checkable physics lives in the closed-form energy ($4\pi J$), the topological invariant (Berg–Lüscher Q), and the exponent algebra — none of which require the expensive coupled solver. Two points of note:

- **No DMI.** The task brief used “DMI” generically; the paper contains *no* Dzyaloshinskii–Moriya term. Its skyrmion is the pure-exchange BP soliton, stabilized dynamically by the current, not energetically by DMI. We replicated what the paper does and did not bolt on a DMI term.
- **Finite-box falloff is not disagreement.** The $E/(4\pi J)$ drop at $\lambda = 64$ on $L = 512$ is the skyrmion tail reaching the open boundary, not a failure of the $4\pi J$ result; the $\lambda = 8$ –16 window is the clean comparison.

5 Verdict

PARTIAL — Coverage 6/10, Agreement 8/10. Replicated: BP exchange energy $4\pi J$, quantized topological charge $Q = \pm 1$ with correct sign, lattice scale-invariance breaking driving barrier-free collapse, and the self-similar exponents $\alpha = 1$, $\beta = 1/2$. Not replicated (honest gaps): the coupled Schrödinger+Landau–Lifshitz time integration with itinerant electrons, the collapse times t^* , and the electron b-field topological-vortex nucleation. See `report/failure_analysis.md` and `report/open_questions.json`.